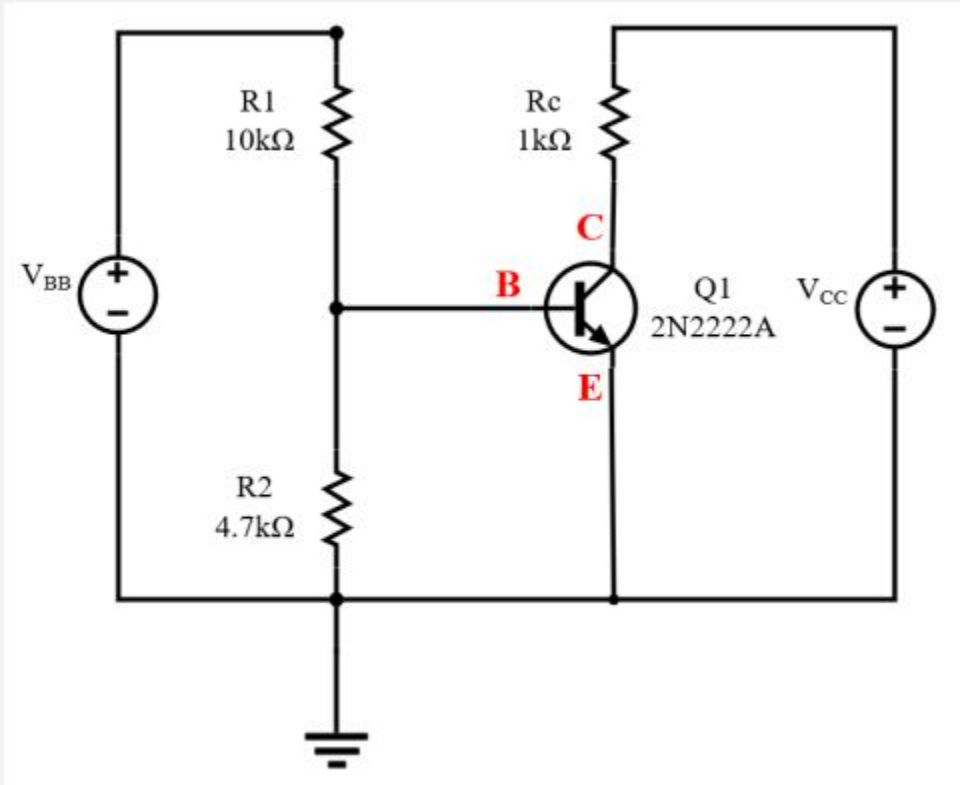
Lab 2 實驗量測項目


R1
10kΩ
Rc
1kΩ
C
B
E
Q1
2N2222A
V_{BB}
V_{CC}
R2
4.7kΩ

Procedure

2. 畫 BJT 特性曲線 IC - VCE curve

- Consider the circuit of Fig 4. using V_{CC} , R_1 = 10 kΩ, R_2 = 4.7 kΩ, and R_C = 1 kΩ.
- Set different V_{BE} and V_{CE} values to measure the collector current I_C, and plot the BJT characteristic curves.

1. 量測數據

VCC=0V, 0.5V, 1V, ..., 15V

記得要量 VBE

V_{BB} = 1.5 V、V_{BE} = V、ON/OFF	
V_{CE} (experimental setting)	I_C (measure) $(=\frac{V_C}{R_C})$ 紀錄 BJT 是 ON/OFF
V	A 量測 IC
V 每個 VCC 下要量測各自的 VCE	A
V	A 可用 C 端電阻的 VC/VR 求得 IC
V	A
V	A

記得要量 VBE

VCC=0V, 0.5V, 1V, ..., 15V

V_{BB} = 2.2 V、V_{BE} = V、ON/OFF	
V_{CE} (experimental setting)	I_C (measure) $(=\frac{V_C}{R_C})$ 紀錄 BJT 是 ON/OFF
V	A 量測 IC
V 每個 VCC 下要量測各自的 VCE	A
V	A 可用 C 端電阻的 VC/VR 求得 IC
V	A
V	A

VCC=0V, 0.5V, 1V, ..., 15V

記得要量 VBE

V_{BB} = 2 V、V_{BE} = V、ON/OFF	
V_{CE} (experimental setting)	I_C (measure) $(=\frac{V_C}{R_C})$ 紀錄 BJT 是 ON/OFF
V	A 量測 IC
V 每個 VCC 下要量測各自的 VCE	A
V	A 可用 C 端電阻的 VC/VR 求得 IC
V	A
V	A

記得要量 VBE

VCC=0V, 0.5V, 1V, ..., 15V

V_{BB} = 2.4 V、V_{BE} = V、ON/OFF	
I_C (measure) $(=\frac{V_C}{R_C})$	I_C (measure) $(=\frac{V_C}{R_C})$ 紀錄 BJT 是 ON/OFF
V	A 量測 IC
V 每個 VCC 下要量測各自的 VCE	A
V	A 可用 C 端電阻的 VC/VR 求得 IC
V	A
V	A

Table 1

Procedure

- Record the results in the first row of Table 3. Based on these data, calculate and record the experimentally obtained β (current gain).

	I_B	I_C	$\beta_{Calculate}(I_C/I_B)$	β_{Meas}
BJT				

X
應該是沒有這一項

Table 2

VBB=?

VCC=?

RC=?

R1=?

R2=?